

INFINITELY MANY SUPERSINGULAR PRIMES FOR SOME ABELIAN VARIETIES OVER REAL NUMBER FIELDS

FANGU CHEN
fangu@berkeley.edu

ABSTRACT. Baba and Granath generalized Elkies' theorem on infinitude of supersingular primes for elliptic curves to abelian surfaces with quaternionic multiplication of discriminant 6, whose field of moduli is $\mathbb{Q}$ and which are Jacobians in characteristic 2 and 3. We extend their result to arbitrary fields of moduli admitting a real embedding and weaken the local conditions at 2 and 3. We also prove, under explicit local hypotheses, the infinitude of supersingular primes for certain Kuga–Satake abelian varieties, extending results previously established in the good reduction setting to a Shimura curve with bad reduction at a finite place. The proofs use CM cycles on Shimura curves, together with arithmetic intersections, deformation theory, and Čerednik–Drinfeld uniformization at the bad primes.

1. INTRODUCTION

1.1. Background. Elkies proved that an elliptic curve defined over a number field of odd degree has infinitely many primes of supersingular reduction [Elk87], and extended the result to elliptic curves over any number field with at least one real embedding [Elk89]. This has been generalized to various families of abelian varieties, including abelian surfaces with multiplication by the quaternion algebra of discriminant 6, with field of moduli $\mathbb{Q}$ and which are Jacobians in characteristic 2 and 3 [BG08b], and certain Kuga–Satake abelian varieties constructed from K3-type Hodge structures with real multiplication by F on the trace zero part of a quaternion algebra over F [Che25]. For a detailed discussion of related works, see [Che25].

1.2. The main results.

1.2.1. We extend the result of [BG08b] to number fields with at least one real embedding, and weaken the local conditions at the bad reduction primes 2 and 3. A special case of our main theorem is as follows.

Theorem 1.1. *Let C be a genus 2 curve with Jacobian that has multiplication by the maximal quaternion order with discriminant 6, and has field of moduli a number field L with at least one real embedding. Assume C has potentially smooth stable reduction at primes above 2 and 3. Then its Jacobian has superspecial reduction at infinitely many primes.*

More generally, the coarse moduli variety of principally polarized abelian surfaces with potential multiplication by the maximal quaternion order of discriminant 6 is isomorphic to $\mathbb{P}^1$, and an arithmetic j -function (see (3.1.1) in §3.1) is defined in [BG08a]. In terms of this coordinate, the assumption at primes $\mathfrak{p}$ above 2 and 3 in Theorem 1.1 corresponds to the case $v_{\mathfrak{p}}(j(C)) = 0$ by [BG08b, Proposition 2], and our main theorem can be stated as follows.

Theorem 1.2. *Let C be a genus 2 curve with Jacobian that has multiplication by the maximal quaternion order with discriminant 6, and has field of moduli a number field L with at least one real embedding. Write $j_0 := j(C) \in L$ and $\mathrm{Nm}_{L/\mathbb{Q}}(j_0) = \frac{n}{d}$ with $n, d \in \mathbb{Z}$, $(n, d, 6) = 1$, $d > 0$. Assume at least one of the following conditions:*

- (1) $v_{\mathfrak{p}}(j_0) \leq 0$ for $\mathfrak{p}$ above 2, 3, and $[L : \mathbb{Q}] + v_3(d \mathrm{Nm}_{L/\mathbb{Q}}(27j_0 + 16))$ is odd or j_0 has at least one real conjugate in $(-\frac{16}{27}, 0)$;
- (2) $v_{\mathfrak{p}}(j_0) \geq 0$ for $\mathfrak{p}$ above 2, $v_{\mathfrak{p}}(j_0) \leq 0$ for $\mathfrak{p}$ above 3, and j_0 has at least one real conjugate in $(0, \infty)$;
- (3) $[L : \mathbb{Q}]$ is even, $v_{\mathfrak{p}}(j_0) \geq 0$ for $\mathfrak{p}$ above 2, 3, and j_0 has at least one real conjugate in $(-\infty, -\frac{16}{27}) \cup (0, \infty)$.

Then the Jacobian of C has superspecial reduction at infinitely many primes.

1.2.2. We extend the result of [Che25] to a family parametrized by a quaternionic Shimura curve over the real quadratic field $F = \mathbb{Q}(\sqrt{5})$, allowing the defining quaternion algebra to ramify at the finite prime $\mathfrak{p}_2$.

Theorem 1.3. *Let $F = \mathbb{Q}(\sqrt{5})$ and B be the quaternion algebra over F ramified exactly at $\mathfrak{p}_2$ and at one real place of F . Let A be a Kuga–Satake abelian variety constructed as in [Che25] from a $K3$ -type Hodge structure with real multiplication by $F = \mathbb{Q}(\sqrt{5})$ on the trace zero part of B . If A gives an F -point on the Shimura curve $\text{Sh}_{K_0}(G_0, \mathcal{H}^\pm)$ defined in [Che25, 2.1.4] and satisfies the local conditions 4.4.3 and 4.4.2, then A has supersingular reduction at infinitely many primes.*

Remark 1.4. This is the case treated by [vG08, Proposition 5.7]. A generic Kuga–Satake abelian variety A in the theorem is isogenous to A_4^8 , where A_4 is a simple abelian fourfold whose endomorphism algebra $\text{End}^0(A_4)$ is the definite quaternion algebra over $\mathbb{Q}$ ramified at 2. Consequently, the theorem also proves the infinitude of supersingular primes for the corresponding abelian fourfolds with definite quaternionic multiplication.

1.3. The proof ideas. Both proofs build on Elkies’ strategy for producing supersingular primes through intersections with suitably chosen CM cycles on a Shimura curve. A common ingredient is that the relevant Shimura curve admits a model isomorphic to $\mathbb{P}^1$ over a totally real number field F , so that we may choose a coordinate j . For an appropriate CM extension $L_\lambda = F(\sqrt{-\lambda})$ of F , we construct a CM cycle $\mathcal{P}_\lambda$ and its associated polynomial $P_\lambda(x) = \prod_{a \in \mathcal{P}_\lambda} (x - j(a)) \in F[x]$. After clearing denominators, the intersection between the point x_0 corresponding to the fixed abelian variety and the CM cycle $\mathcal{P}_\lambda$ at a prime $\mathfrak{p}$ is detected by

$$v_{\mathfrak{p}}(P_\lambda(j(x_0))) > 0.$$

When such an intersection occurs, the condition

$$\left(\frac{-\lambda}{\mathfrak{p}}\right) \neq 1$$

implies that x_0 has supersingular reduction at $\mathfrak{p}$. To produce such a prime $\mathfrak{p}$, we must therefore understand both the reduction of the CM cycle $\mathcal{P}_\lambda$ modulo primes dividing λ and the signs of the real conjugates of $P_\lambda(j(x_0))$.

At a good reduction prime that ramifies in L_λ , a deformation-theoretic method shows that the reductions of the CM points in $\mathcal{P}_\lambda$ typically occur in pairs, apart from a small and explicitly determined set of exceptional points with extra automorphisms. By studying the endomorphism algebra, or the space of special endomorphisms, of a point in the special fiber, we can use suitable congruence conditions on λ to rule out some of these exceptional points.

The signs of the real conjugates of $P_\lambda(j(x_0))$ are determined by the positions of the real roots of P_λ relative to $j(x_0)$. The real locus of the Shimura curve is the image of finitely many geodesics in its complex uniformization, and the existence of a CM point in $\mathcal{P}_\lambda$ on a given geodesic is equivalent to the solvability of an associated norm equation. Congruence conditions on λ can be used to select the real components on which the CM points occur. The archimedean hypotheses in the main theorems ensure that at least one real conjugate of x_0 lies on a real component reached by the chosen CM cycles. An equidistribution theorem for primes represented by these norm forms then allows the real CM points to be placed on the required side of that conjugate while retaining the required finite congruence conditions on λ , producing the desired sign in the quadratic reciprocity formula.

To prove infinitude, we begin with a finite set S of primes and construct a new supersingular prime $\mathfrak{p} \notin S$. For every good prime $\mathfrak{q} \in S$, we impose

$$\left(\frac{-\lambda}{\mathfrak{q}}\right) = 1.$$

This condition cannot be imposed when the quaternion algebra is ramified at $\mathfrak{q}$. Instead, we must show directly that x_0 does not meet the chosen CM cycle in the special fiber at such $\mathfrak{q}$. This is the source of the local hypotheses at the bad primes in our theorems. For example, in the discriminant 6 case, [BG08b] shows that, in characteristic 2 and 3, the reduction of any CM cycle lies in the superspecial locus $\{j = 0, \infty\}$, and the intersection formula of Heegner divisors in [KR08] provides the new input that determines the specific superspecial point for each chosen CM cycle. The local conditions on j_0 at primes above 2 and 3 ensure that it avoids intersection with the chosen CM cycles at these primes.

A more detailed, case-by-case study, where in each case λ is chosen in a form adapted to the conditions, allows one to weaken the local conditions and obtain a more general result. We expect this method to apply to other families parametrized by Shimura curves, including variants of the settings considered in [Jao03, Sad04, LMPT25, Che25].

1.4. Notations and conventions. Let F be a totally real number field with narrow class number 1, and B be a quaternion algebra over F split at exactly one real place $\rho : F \hookrightarrow \mathbb{R}$. Let $\mathcal{O} = \mathcal{O}_B$ be a maximal order of B .¹ In the two applications considered, we specialize either to $F = \mathbb{Q}$, with B the indefinite quaternion algebra over $\mathbb{Q}$ of discriminant 6, or to $F = \mathbb{Q}(\sqrt{5})$, with B ramified exactly at $\mathfrak{p}_2$ and at one real place, where $\mathfrak{p}_2$ denotes the unique prime of $\mathbb{Q}(\sqrt{5})$ above 2.

2. PRELIMINARIES

In this section, we recall the setup in [BG08b] and [Che25].

2.1. The Shimura curves. Let $\tilde{G} = \text{Res}_{F/\mathbb{Q}} \text{GL}_{1,B}$ be the algebraic group over $\mathbb{Q}$ with $\tilde{G}(\mathbb{Q}) = B^\times$, and $\tilde{K} = \hat{\mathcal{O}}^\times = \prod \mathcal{O}_v^\times \subset \tilde{G}(\mathbb{A}_f)$. Since F has narrow class number 1, strong approximation implies that any two maximal orders in the indefinite quaternion algebra B are conjugate to each other and the Shimura curve

$$\text{Sh}_{\tilde{K}}(\tilde{G}, \mathcal{H}^\pm) = \tilde{G}(\mathbb{Q}) \backslash \mathcal{H}^\pm \times \tilde{G}(\mathbb{A}_f) / \tilde{K} = \tilde{\Gamma} \backslash \mathcal{H} = \tilde{\Gamma}^1 \backslash \mathcal{H},$$

where

$$\begin{aligned} \tilde{\Gamma} &= \{\alpha \in \mathcal{O}^\times : \text{nrd}(\alpha) \in \mathcal{O}_F^\times \text{ is totally positive}\}, \\ \tilde{\Gamma}^1 &= \mathcal{O}^1 = \{\alpha \in \mathcal{O}^\times : \text{nrd}(\alpha) = 1\}. \end{aligned}$$

The reflex field of this Shimura curve is $\rho(F)$.

Let G_0 be the algebraic group over $\mathbb{Q}$ with $G_0(\mathbb{Q}) = B^\times / F^\times$, and $K_0 \subset G_0(\mathbb{A}_f)$ be defined as in [Che25, 2.1.4]. Let $\tilde{\Gamma}^* = \{\alpha \in N_{B^\times}(\mathcal{O}) : \text{nrd}(\alpha) \text{ is totally positive}\}$. Since F has narrow class number 1,

$$\tilde{\Gamma}^* / F^\times \tilde{\Gamma} = N_{B^\times}(\mathcal{O}) / F^\times \mathcal{O}^\times \simeq \prod_{\text{prime ideal } \mathfrak{p} \mid \text{disc}(B)} (\mathbb{Z}/2\mathbb{Z})$$

is the Atkin–Lehner group $W(\mathcal{O})$, and the Atkin–Lehner involutions give a finite morphism

$$\text{Sh}_{\tilde{K}}(\tilde{G}, \mathcal{H}^\pm) = \tilde{\Gamma} \backslash \mathcal{H} \rightarrow \tilde{\Gamma}^* \backslash \mathcal{H} = \text{Sh}_{K_0}(G_0, \mathcal{H}^\pm)$$

of degree $2^{\#\text{finite primes ramified in } B}$.

2.1.1. The case $F = \mathbb{Q}$. Denote the discriminant of B by Δ . Fix an element $\mu \in \mathcal{O}$ such that $\mu^2 = -\Delta$,² then the involution $\alpha \mapsto \alpha' = \mu^{-1} \bar{\alpha} \mu$ is a positive anti-involution on B . The Shimura curve $V_\Delta := \text{Sh}_{\tilde{K}}(\tilde{G}, \mathcal{H}^\pm) = \tilde{\Gamma}^1 \backslash \mathcal{H}$ is the coarse moduli space of isomorphism classes of $[A, \iota]$, where A is a principally polarized abelian surface, and $\iota : \mathcal{O} \hookrightarrow \text{End}(A)$ is an embedding such that the Rosati involution defined by the polarization on $\iota(\mathcal{O})$ is $'$.³ Fix an isomorphism $\iota_\infty : B_\mathbb{R} \xrightarrow{\sim} M_2(\mathbb{R})$. For any point τ on the upper half plane $\mathcal{H}$, consider the complex torus $A_\tau = \mathbb{C}^2 / \Lambda_\tau$, where

$$\Lambda_\tau = \iota_\infty(\mathcal{O}) \begin{pmatrix} \tau \\ 1 \end{pmatrix}$$

is a lattice in $\mathbb{C}^2$, the map ι_∞ induces a natural embedding $\iota_\tau : \mathcal{O} \hookrightarrow \text{End}(A_\tau)$ as $\mathcal{O}$ is closed under multiplication. After possibly replacing μ by $-\mu$ once and for all, the Riemann form

$$\begin{aligned} \Lambda_\tau \times \Lambda_\tau &\rightarrow \mathbb{Z}, \\ \left(\iota_\infty(x) \begin{pmatrix} \tau \\ 1 \end{pmatrix}, \iota_\infty(y) \begin{pmatrix} \tau \\ 1 \end{pmatrix} \right) &\mapsto \frac{1}{\Delta} \text{trd}(\mu x \bar{y}) \end{aligned}$$

¹If F has narrow class number 1, then any two maximal orders in an indefinite quaternion algebra over F are conjugate to each other.

²The element μ exists because the field $\mathbb{Q}(\sqrt{-\Delta})$ embeds in B by the local-global principle, and any two maximal orders in B are conjugate to each other.

³Given an abelian surface A and $\iota : \mathcal{O} \hookrightarrow \text{End}(A)$, there is a unique principal polarization on A such that the induced Rosati involution is compatible with μ .

defines the unique principal polarization on A_τ compatible with ι_τ . Let Γ^1 be the image in $\mathrm{PSL}_2(\mathbb{R})$ of $\iota_\infty(\mathcal{O}^1)$, then the map $\Gamma^1 \tau \mapsto [A_\tau, \iota_\tau]$ gives a bijection $\Gamma^1 \backslash \mathcal{H} \rightarrow V_\Delta(\mathbb{C})$.

There is the natural forgetful map $V_\Delta \rightarrow \mathcal{A}_2$ from V_Δ to the moduli threefold of principally polarized abelian surfaces under which $[A, \iota]$ maps to $[A, \mathcal{C}]$ where $\mathcal{C}$ is the unique principal polarization on A compatible with ι . Let E_Δ denote its image and E_Δ^0 denote the intersection of E_Δ with the image of the moduli space of genus 2 curves under the open Torelli map.

From now on, let $\Delta = 6$. By [Rot04], the forgetful map $V_6 \rightarrow \mathcal{A}_2$ has degree 4 and factors as $V_6 \rightarrow V_6/W(\mathcal{O}) \simeq E_6 \hookrightarrow \mathcal{A}_2$, where the Atkin–Lehner group $W(\mathcal{O}) = \{1, w_2, w_3, w_6\}$, and w_d is represented by some $\chi_d \in \mathcal{O} \cap N_{B^\times}(\mathcal{O})$ with $\mathrm{trd}(\chi_d) = 0$, $\mathrm{nrd}(\chi_d) = d$. Moreover, there is a unique $N_{B^\times}(\mathcal{O})$ -conjugacy class of embedding $\mathbb{Z}[\sqrt{-6}] \hookrightarrow \mathcal{O}$, then the image E_6 is independent of the choice of μ and E_6 is the moduli space of principally polarized abelian surfaces with potential QM by $\mathcal{O}$.

$$\begin{array}{ccc} & & V_6 \\ & & \downarrow 4 \\ E_6^0 & \longrightarrow & E_6 \\ \downarrow & & \downarrow \\ \mathcal{M}_2 & \longrightarrow & \mathcal{A}_2 \end{array}$$

2.1.2. The case $[F : \mathbb{Q}] > 1$. As in [Che25], we consider the Shimura curve $\mathrm{Sh}_{K_0}(G_0, \mathcal{H}^\pm)$ as a finite quotient of a Hodge-type Shimura curve, which parametrizes isogeny classes of abelian varieties with additional structures. When $F = \mathbb{Q}(\sqrt{5})$ and B is the quaternion algebra over F ramified exactly at $\mathfrak{p}_2$ and at one real place of F , the Shimura curve $\tilde{\Gamma} \backslash \mathcal{H} = \mathrm{Sh}_{\tilde{K}}(\tilde{G}, \mathcal{H}^\pm)$ has signature $(0; 2, 5, 5)$ ([Voi09]) with an F -rational point given by the elliptic point of order 2, and the Atkin–Lehner involution gives a degree 2 morphism $\mathrm{Sh}_{\tilde{K}}(\tilde{G}, \mathcal{H}^\pm) \rightarrow \mathrm{Sh}_{K_0}(G_0, \mathcal{H}^\pm)$. At an odd prime p , the image of $\tilde{K}_p$ is $K_{0,p}$. The Shimura curve $\mathrm{Sh}_{K_0}(G_0, \mathcal{H}^\pm)$ has signature $(0; 2, 4, 5)$ ([Tak77]) and canonical model $\mathbb{P}_F^1$.

2.2. CM points. For $\lambda \in \mathcal{O}_F$ totally positive, put $L_\lambda = F(\sqrt{-\lambda})$. The new local condition for L_λ to embed into B is given by

$$-\lambda \notin (F_{\mathfrak{p}}^\times)^2$$

for any prime $\mathfrak{p}$ at which $B_{\mathfrak{p}}$ is division. Fix an embedding $\phi : L_\lambda \hookrightarrow B$ with $\phi(\mathcal{O}_{L_\lambda}) \subset \mathcal{O}$, i.e., ϕ defines an optimal embedding $\mathcal{O}_{L_\lambda} \hookrightarrow \mathcal{O}$.

Remark 2.1. More generally, we can consider optimal embeddings $\mathcal{R} \hookrightarrow \mathcal{O}$, where $\mathcal{R}$ is an order in L_λ containing $\mathcal{O}_F[\sqrt{-\lambda}]$. Note that at a prime $\mathfrak{p}$ dividing $\mathrm{disc}(B)$, optimality forces $L_{\lambda, \mathfrak{p}} \cap \mathcal{O}_{\mathfrak{p}} = \mathcal{O}_{L_\lambda, \mathfrak{p}}$; hence $\mathfrak{p}$ does not divide the conductor of $\mathcal{R} \subseteq \mathcal{O}_{L_\lambda}$.

The formal construction of the CM cycle from the torus

$$\tilde{T}_\lambda = \mathrm{Res}_{L_\lambda/\mathbb{Q}} \mathbb{G}_m$$

and a cocharacter $\tilde{h}$ is the same as in [Che25, §3.1], where Lemmas 3.1 and 3.2 generalize to the present setting as follows.

Lemma 2.2. *Let $\hat{\mathcal{E}} := \{\hat{\beta} \in \hat{B}^\times : \hat{\beta}^{-1} \widehat{\phi(L_\lambda)} \hat{\beta} \cap \hat{\mathcal{O}} = \hat{\beta}^{-1} \phi(\widehat{\phi^{-1}(\mathcal{O})}) \hat{\beta}\}$.*

- (1) *When $\hat{B}^\times = B^\times \hat{\mathcal{O}}^\times$, the set of CM points $\{[\tilde{h}, e] : e \in \hat{\mathcal{E}}\}$ on $\tilde{\Gamma} \backslash \mathcal{H}$ corresponds to $\mathcal{O}^\times$ -conjugacy classes of optimal embeddings $\mathcal{O}_{L_\lambda} \hookrightarrow \mathcal{O}$.*
- (2) *The set of CM points $\{[\tilde{h}, e] : e \in \hat{\mathcal{E}}\}$ on $\tilde{\Gamma} \backslash \mathcal{H}$ has cardinality*

$$h(L_\lambda) \prod_{\mathfrak{p} \mid \mathrm{discrd} B} \left(1 - \left(\frac{L_\lambda}{\mathfrak{p}}\right)\right),$$

and is a union of $\prod_{\mathfrak{p} \mid \mathrm{discrd} B} \left(1 - \left(\frac{L_\lambda}{\mathfrak{p}}\right)\right)$ CM cycles.

Lemma 2.3. *Complex conjugation defines an involution on the set of CM points $\{[\tilde{h}, e] : e \in \hat{\mathcal{E}}\}$.*

2.2.1. *Examples when $F = \mathbb{Q}$ and B ramified at 2, 3.* Let $\mathcal{O}_D$ be an imaginary quadratic order of discriminant D such that $\mathcal{O}_D \hookrightarrow \mathcal{O}$ is an optimal embedding. The fixed point τ of $\iota_\infty(\mathcal{O}_D)$ in $\mathcal{H}$ is a CM point, and its corresponding abelian variety A_τ has $\text{End}_{\text{QM}}(\mathcal{A}_\tau) \simeq \mathcal{O}_D$. From now on, let $D < 0$ be a fundamental discriminant such that $\mathcal{O}_D \hookrightarrow \mathcal{O}$ and $E_6(D)$ denote the set of points with CM by $\mathcal{O}_D$. Let W'' denote the subgroup of the Atkin–Lehner group generated by elements w_p , where $p|\Delta$ is a prime ramified in $\mathbb{Q}(\sqrt{D})$, then the number of elements in $E_6(D)$, counted with appropriate multiplicities, is

$$h' = \frac{h(\mathcal{O}_D)}{\#W''}.$$

By genus theory and [CH88, (19.6)], the parity of h' can be determined. Let l be a prime. The relevant cases considered in this work are summarized in the following table.

D	$h(\mathcal{O}_D)$	h'
$-4l, l \equiv 13 \pmod{24}$	$\equiv 2 \pmod{4}$	odd
$-l, l \equiv 19 \pmod{24}$	odd	odd
$-3l, l \equiv 1 \pmod{24}$	$\equiv 0 \pmod{4}$	even

2.2.2. *Examples when $F = \mathbb{Q}(\sqrt{5})$ and B ramified at $\mathfrak{p}_2$.* The elliptic points on $\text{Sh}_{\tilde{K}}(\tilde{G}, \mathcal{H}^\pm)$ are CM points, where the order-2 point corresponds to $\mathcal{O}_F[\sqrt{-1}] \hookrightarrow \mathcal{O}$ and the two order-5 points correspond to $\mathcal{O}_F[\zeta_5] \hookrightarrow \mathcal{O}$.

The $\mathfrak{p}_2$ -Atkin–Lehner involution acts on the set of CM points $\{[\tilde{h}, e] : e \in \hat{\mathcal{E}}\}$. It fixes two points: the unique order-2 elliptic point corresponding to $\mathcal{O}_F[\sqrt{-1}] \hookrightarrow \mathcal{O}$ and a second point corresponding to $\mathcal{O}_F[\sqrt{-2}] \hookrightarrow \mathcal{O}$; it exchanges the two order-5 elliptic points. Since $\mathfrak{p}_2$ is the only finite prime ramifying in B , the number of images of those CM points on $\text{Sh}_{K_0}(G_0, \mathcal{H}^\pm)$ is $h(L_\lambda)$ if $\mathfrak{p}_2$ is inert in L_λ . In particular, if λ is a totally positive prime of F satisfying

$$(2.2.1) \quad -\lambda \equiv \square \pmod{4}, \quad -\lambda \not\equiv \square \pmod{8},$$

then $\mathfrak{p}_2$ is inert in L_λ and $h(L_\lambda)$ is odd by [CH88, (13.7)].

3. PROOF OF THEOREM 1.2

In this section, we prove Theorem 1.2 by applying the method of §1.3 to three families of Heegner cycles. In §3.1, we recall the arithmetic j -coordinate on E_6 and translate the results of [Jao03] into this coordinate; these results describe both the reductions of the associated Heegner polynomials modulo l and the locations of their real roots. In §3.2, we use the arithmetic intersection formula of [KR08] to determine their reductions at the bad primes 2 and 3. Finally, in §3.3, we combine these results to produce a supersingular, and hence superspecial, prime outside any prescribed finite set.

3.1. The coordinate functions. It is shown in [BG08a, 3.6] that there is an isomorphism $j = j_6 : E_6^0 \rightarrow \mathbb{P}^1 \setminus \{0, \infty\}$ given by

$$(3.1.1) \quad j = \frac{12^{10} J_{10}^2}{(J_2^2 - 24J_4)^5}.$$

Denote the elements of $E_6(D)$ to be $a_1, \dots, a_{h'}$, and define

$$P_D(x) = b_{h'} \prod_{i=1}^{h'} (x - j(a_i)),$$

where $b_{h'} > 0$ is the smallest integer such that $P_D(x) \in \mathbb{Z}[x]$.

Elkies ([Elk98]) computed a different rational coordinate function $t : E_6 \rightarrow \mathbb{P}^1$, which is used in [Jao03]. The relation between the two coordinate functions is $j = 16(t - 1)/27$. In particular, from the following correspondence of the coordinates of the elliptic points we can translate the results in [Jao03] to our setting.

Elliptic point of order	CM	j	t
6	$\mathbb{Z} \left[\frac{1+\sqrt{-3}}{2} \right]$	∞	∞
4	$\mathbb{Z} [\sqrt{-1}]$	0	1
2	$\mathbb{Z} [\sqrt{-6}]$	$-\frac{16}{27}$	0

Lemma 3.1 ([Jao03, Theorems 3.4.1 and 3.4.2]). *Let $l \geq 5$ be a prime. For discriminants D of the form in the table, each root of $P_D(x) \bmod l$ occurs with even multiplicity, except possibly for roots corresponding to points on E_6 which are congruent modulo l to one of the three elliptic points. The divisor of unpaired zeros of $P_D(x)$ modulo l is as follows: (see [Jao03, Theorem 3.4.2] for a complete list)*

D	divisor
$-4l, l \equiv 13 \bmod 24$	$(-16/27)$
$-l, l \equiv 19 \bmod 24$	$(-16/27)$
$-3l, l \equiv 1 \bmod 24$	$\emptyset$

Lemma 3.2 ([Jao03, Theorem 3.3.1]). *Let $l \geq 5$ be a prime. For discriminants D of the form in the table, the polynomial $P_D(x)$ has at most one real root in each of the intervals $I_1 = (-\infty, -\frac{16}{27})$, $I_2 = (-\frac{16}{27}, 0)$, and $I_3 = (0, \infty)$, with the roots being located as follows: (see [Jao03, Theorem 3.3.1] for a complete list, $*$ denotes the presence of a real root in the interval)*

D	I_1	I_2	I_3
$-4l, l \equiv 13 \bmod 24$		*	
$-l, l \equiv 19 \bmod 24$			*
$-3l, l \equiv 1 \bmod 24$	*		*

Furthermore, for any subinterval and sub-congruence class of any starred entry, there exist infinitely many primes l in that sub-congruence class for which $P_D(x)$ has a real root in the subinterval.

Remark 3.3. Let $F_1 = \mathbb{Q}(\sqrt{2})$, $F_2 = \mathbb{Q}(\sqrt{3})$, $F_3 = \mathbb{Q}(\sqrt{6})$, and ε_i be a fundamental unit of F_i , $i = 1, 2, 3$. The claim that any subinterval of an interval I_i , in which polynomials P_D with D of the specified form have real roots, supports infinitely many such D follows from an equidistribution result ([Lan94, XV, §5]) for primes in F_i , by constructing a Hecke character $\sigma_i : \mathbb{A}_{F_i}^\times \rightarrow \mathbb{R}/(\ln \varepsilon_i)\mathbb{Z}$ given by

$$\mathbb{A}_{F_i}^\times / (F_i^\times \prod_{\mathfrak{p} \nmid \infty} U_{\mathfrak{p}}) \simeq (\mathbb{R}^\times \times \mathbb{R}^\times) / U_{F_i} \rightarrow \mathbb{R}/(\ln \varepsilon_i)\mathbb{Z},$$

$$(a, b) \mapsto \frac{1}{2} \ln \frac{|a|}{|b|}.$$

If the polynomials P_D have real roots in both I_{i_1} and I_{i_2} , then the result generalizes to any open subset of $I_{i_1} \times I_{i_2}$ by equidistribution of primes in the compositum $F_{i_1} F_{i_2}$. For example, let $F = F_1 F_3$ and define $\sigma = (\sigma_1 \circ \text{Nm}_{F/F_1}, \sigma_3 \circ \text{Nm}_{F/F_3}) : \mathbb{A}_F^\times \rightarrow \mathbb{R}/(\ln \varepsilon_1)\mathbb{Z} \times \mathbb{R}/(\ln \varepsilon_3)\mathbb{Z}$, then $\sigma(a, b, c, d) = (\frac{1}{2} \ln \frac{|ac|}{|bd|}, \frac{1}{2} \ln \frac{|ab|}{|cd|})$ for $(a, b, c, d) \in \mathbb{R}^\times \times \mathbb{R}^\times \times \mathbb{R}^\times \times \mathbb{R}^\times \hookrightarrow \mathbb{A}_F^\times$. In particular, $\sigma(a, \frac{1}{a}, 1, 1) = (\ln |a|, 0)$ and $\sigma(a, 1, \frac{1}{a}, 1) = (0, \ln |a|)$, so the restriction of σ to the subgroup of ideles of norm 1 is surjective. Let $\tau : \{\text{primes of } F\} \rightarrow \mathbb{A}_F^\times$ be the map taking a prime $\mathfrak{p}$ to an idele that is 1 at all places except a prime element of $F_{\mathfrak{p}}$ at $\mathfrak{p}$, and $\Lambda = \sigma \circ \tau$, then the set of primes of F is Λ -equidistributed in $\mathbb{R}/(\ln \varepsilon_1)\mathbb{Z} \times \mathbb{R}/(\ln \varepsilon_3)\mathbb{Z}$. Since a density 1 set of primes of F lies above a totally split rational prime, the set

$$\left\{ \left(\frac{1}{2} \ln \left| \frac{\pi_1}{\pi'_1} \right|, \frac{1}{2} \ln \left| \frac{\pi_3}{\pi'_3} \right| \right) : l \text{ rational prime, } (l) = (\pi_1)(\pi'_1) \text{ in } F_1, (l) = (\pi_3)(\pi'_3) \text{ in } F_3 \right\}$$

is equidistributed in $\mathbb{R}/(\ln \varepsilon_1)\mathbb{Z} \times \mathbb{R}/(\ln \varepsilon_3)\mathbb{Z}$.

3.2. Integral model and intersection number at bad primes. In this subsection, we compute $P_D(x)$ modulo 2 and 3 by computing intersection numbers of Heegner divisors on the integral model of E_6 .

Let $\mathcal{X} = \mathcal{X}_{1,6,m}$ be the moduli space for abelian surfaces with additional structure as in [KR08, §2]. An integral model $\mathcal{E}$ of E_6 (resp. $\mathcal{V}$ of V_6) is obtained as the quotient of $\mathcal{X}$ by a finite subgroup of order $4 \cdot 2\eta(m)$ (resp. $2\eta(m)$), where $\eta(m)$ is defined in [KR08, (3.4)].

Let p be a prime. For $p \neq 2, 3$, the curve E_6 has good reduction at p , and $\mathcal{E}_{\mathbb{F}_p} \simeq \mathbb{P}_{\mathbb{F}_p}^1$. Suppose $p|\Delta$. As a consequence of the theorem of Čerednik–Drinfeld (see for instance [BC91]), the formal completion of the integral model along its special fiber at p is isomorphic to a finite union of Galois twists (over unramified extensions) of quotients of Mumford curves ([Mum72]). Then the geometric special fiber $\mathcal{E}_{\overline{\mathbb{F}_p}}$ of $\mathcal{E}$ at p can be viewed as a graph, and we follow [Kur79] to compute its dual graph and thus determine $\mathcal{E}_{\mathbb{F}_p}$.

Lemma 3.4. *For $p = 2, 3$, the special fiber $\mathcal{E}_{\mathbb{F}_p} \simeq \mathbb{P}_{\mathbb{F}_p}^1$.*

Proof. Let $B' = B_{\Delta/p}$ be the definite quaternion algebra of discriminant $\frac{\Delta}{p}$, and $\mathcal{O}'$ be a maximal order of B' , so that $\mathcal{O}' \otimes \mathbb{Z}_q \simeq \mathcal{O} \otimes \mathbb{Z}_q$ for any prime $q \neq p$. Let

$$\begin{aligned}\Gamma'_0 &= \mathcal{O}'[p^{-1}]^\times / \mathbb{Z}[p^{-1}]^\times, \\ \Gamma'^* &= \{\gamma \in B'^\times : \gamma \mathcal{O}'[p^{-1}] = \mathcal{O}'[p^{-1}]\gamma\}\end{aligned}$$

regarded as discrete subgroups of $\mathrm{PGL}_2(\mathbb{Q}_p)$, and $\mathcal{T}$ denote the Bruhat–Tits tree. We have graphs with lengths $\Gamma'_0 \backslash \mathcal{T}$ and $\Gamma'^* \backslash \mathcal{T}$ defined by the quotient of the Bruhat–Tits tree [Kur79, §3]. The number of vertices of $\Gamma'_0 \backslash \mathcal{T}$ equals the class number $h(B')$ of B' [Kur79, p.291].

As explained in detail in [Kur79], the dual graph of the geometric special fiber $\mathcal{E}_{\overline{\mathbb{F}_p}}$ is obtained from $\Gamma'^* \backslash \mathcal{T}$ by removing self-inverse edges. The graph $\Gamma'^* \backslash \mathcal{T}$ is a quotient of $\Gamma'_0 \backslash \mathcal{T}$, and we have $h(B_2) = 1$ and $h(B_3) = 1$, so both $\mathcal{E}_{\overline{\mathbb{F}_2}}$ and $\mathcal{E}_{\overline{\mathbb{F}_3}}$ consist of a single irreducible component. Since the generic fiber $E_6 \simeq \mathbb{P}_{\mathbb{Q}}^1$, which has genus 0 and a rational point, it follows that $\mathcal{E}_{\mathbb{F}_2} \simeq \mathbb{P}_{\mathbb{F}_2}^1$ and $\mathcal{E}_{\mathbb{F}_3} \simeq \mathbb{P}_{\mathbb{F}_3}^1$. $\square$

Consider $\mathcal{X} \xrightarrow{2\eta(m)} \mathcal{V} \xrightarrow{4} \mathcal{E}$. We will use [KR08, (3.12)] to compute the arithmetic intersection number. Let D be a fundamental discriminant, and $\mathcal{P}_{D,\mathcal{X}}$ be the Heegner divisor of discriminant D on $\mathcal{X}$ as defined in [KR08, §2] and $\mathcal{P}_{D,\mathcal{E}} = \sum_{x \in E_6(D)} \underline{x}$ be the divisor on $\mathcal{E}$. On V_6 , the universal automorphism group is $\{\pm 1\}$, the elliptic points of order 3 have CM by $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$ and the elliptic points of order 2 have CM by $\mathbb{Z}[\sqrt{-1}]$. The Atkin–Lehner involution w_2 has two fixed points (CM by $\mathbb{Z}[\sqrt{-1}]$), w_3 has two fixed points (CM by $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$), w_6 has two fixed points (CM by $\mathbb{Z}[\sqrt{-6}]$). For discriminants $D < -6$, we have

$$(3.2.1) \quad \langle \mathcal{P}_{-3,\mathcal{E}}, \mathcal{P}_{D,\mathcal{E}} \rangle_{\mathcal{E}} = \frac{2}{4} \cdot \frac{2 \cdot 3}{2\eta(m)} \langle \mathcal{P}_{-3,\mathcal{X}}, \mathcal{P}_{D,\mathcal{X}} \rangle_{\mathcal{X}},$$

$$(3.2.2) \quad \langle \mathcal{P}_{-4,\mathcal{E}}, \mathcal{P}_{D,\mathcal{E}} \rangle_{\mathcal{E}} = \frac{2}{4} \cdot \frac{2 \cdot 2}{2\eta(m)} \langle \mathcal{P}_{-4,\mathcal{X}}, \mathcal{P}_{D,\mathcal{X}} \rangle_{\mathcal{X}}.$$

Recall [BG08b, Lemma 3] that $P_D(x) \equiv x^n \pmod{2}$ for some n and $P_D(x) \equiv \pm x^m \pmod{3}$ for some m . For discriminants D considered in this work, we obtain a strengthened version by computing the local intersection number of $\mathcal{P}_{-3,\mathcal{X}}$ or $\mathcal{P}_{-4,\mathcal{X}}$ with $\mathcal{P}_{D,\mathcal{X}}$ at 2 and 3.

Lemma 3.5. *Let D be a fundamental discriminant.*

- (1) *When $3 \nmid D$, the local intersection of $\mathcal{P}_{-3,\mathcal{E}}$ and $\mathcal{P}_{D,\mathcal{E}}$ at p is 0 if $-3D$ is not a square modulo $24p$.*
- (2) *When $2 \nmid D$, the local intersection of $\mathcal{P}_{-4,\mathcal{E}}$ and $\mathcal{P}_{D,\mathcal{E}}$ at p is 0 if $-4D$ is not a square modulo $24p$.*

Proof. This follows immediately from the intersection formula in [KR08, Theorem 3.2] and (3.2.1), (3.2.2). $\square$

Lemma 3.6. *Let $l \geq 5$ be a prime. For discriminants D of the form in the table, the following reductions hold:*

D	$P_D \pmod{2}$	$P_D \pmod{3}$
$-4l, l \equiv 13 \pmod{24}$	$x^{h'}$	$\pm x^{h'}$
$-l, l \equiv 19 \pmod{24}$	1	$\pm x^{h'}$
$-3l, l \equiv 1 \pmod{24}$	1	± 1

Proof. For $D = -4l$ where $l \equiv 13 \pmod{24}$ or $D = -l$ where $l \equiv 19 \pmod{24}$, since $-3D$ is not a square modulo 72, the local intersection of $\mathcal{P}_{-3,\mathcal{E}}$ and $\mathcal{P}_{D,\mathcal{E}}$ at $p = 3$ is 0, then $v_3(j(a)) > 0$ for every root $j(a)$ of $P_D(x)$,

so $P_D(x) \equiv \pm x^{h'} \pmod{3}$. For $l \equiv 1 \pmod{24}$, since $12l$ is not a square modulo 72 , the local intersection of $\mathcal{P}_{-4,\varepsilon}$ and $\mathcal{P}_{-3l,\varepsilon}$ at $p = 3$ is 0 , then $v_3(j(a)) < 0$ for every root $j(a)$ of $P_{-3l}(x)$, so $P_{-3l}(x) \equiv \pm 1 \pmod{3}$.

For $D = -l$ where $l \equiv 19 \pmod{24}$ or $D = -3l$ where $l \equiv 1 \pmod{24}$, since $-4D$ is not a square modulo 48 , the local intersection of $\mathcal{P}_{-4,\varepsilon}$ and $\mathcal{P}_{D,\varepsilon}$ at $p = 2$ is 0 , then $v_2(j(a)) < 0$ for every root $j(a)$ of $P_D(x)$, so $P_D(x) \equiv 1 \pmod{2}$. The case of $P_{-4l}(x) \pmod{2}$ where $l \equiv 13 \pmod{24}$ is proved in [BG08b, Lemma 6]. $\square$

3.3. Proof of main theorem. An abelian surface with action of a maximal order in a rational quaternion algebra B has either ordinary or superspecial reduction modulo primes not dividing the discriminant of B ([Cla03, p.70] and [Rib89, p.23]). It therefore suffices to construct primes of supersingular reduction, for which we follow Elkies' general strategy.

The result is known when the Jacobian of C is a CM QM abelian surface, so we may assume that j_0 is a non-CM point on the Shimura curve. Let S be a finite set of primes containing primes above $2, 3$ and all primes occurring in j_0 and $27j_0 + 16$. Write $P = |\mathrm{Nm}_{L/\mathbb{Q}}(P_D(j_0))|$, $Q = |\mathrm{Nm}_{L/\mathbb{Q}}(27j_0 + 16)|$, $s = \mathrm{sgn}(\mathrm{Nm}_{L/\mathbb{Q}}(P_D(j_0)))$, $s' = \mathrm{sgn}(\mathrm{Nm}_{L/\mathbb{Q}}(27j_0 + 16))$, and $N = d^{h'} P$ where $h' = \deg P_D$. Let $j_1 < j_2 < \dots < j_r$ be the real conjugates of j_0 .

In each case, we are going to find some discriminant D with a rational prime $p \notin \mathrm{Nm}_{L/\mathbb{Q}}(S)$ such that $v_p(P_D(j_0)) > 0$ and $\left(\frac{D}{p}\right) \neq 1$, then as in [BG08b], there is a prime $\mathfrak{p} \notin S$ above p at which the Jacobian of C has supersingular, and hence superspecial reduction.

(1) Choose a prime l satisfying the conditions:

- $l \equiv 13 \pmod{24}$,
- $\left(\frac{-l}{q}\right) = 1$ for every prime $q \in \mathrm{Nm}_{L/\mathbb{Q}}(S) \setminus \{2, 3\}$,
- $P_{-4l}(x)$ has a real root $j(a_i)$ in a subinterval of $(-\frac{16}{27}, 0)$ to be specified later.

Let $D = -4l$. By [BG08b, Lemma 6] and Lemma 3.6,

$$\begin{aligned} P_{-4l}(x) &\equiv x^{h'} \pmod{4}, \\ P_{-4l}(x) &\equiv \pm x^{h'} \pmod{3}, \\ P_{-4l}(x) &\equiv (27x + 16)S(x)^2 \pmod{l} \text{ for some } S(x) \in \mathbb{Z}[x]. \end{aligned}$$

Since $v_{\mathfrak{p}}(j_0) \leq 0$ for all $\mathfrak{p}$ above $2, 3$, we can choose d such that $d \prod_{\sigma \in T} \sigma(j_0)$ is integral for any $T \subset \mathrm{Hom}(L, \overline{\mathbb{Q}})$ and any prime dividing d lies in $\mathrm{Nm}_{L/\mathbb{Q}}(S)$, then $N, dQ \in \mathbb{N}$ and $(N, 6) = 1$, $(dQ, 2) = 1$. Suppose $l \nmid N$, then

$$\begin{aligned} \left(\frac{-4l}{N}\right) &= \left(\frac{-1}{N}\right) \left(\frac{l}{N}\right) = \left(\frac{-1}{N}\right) \left(\frac{N}{l}\right) \\ &= \left(\frac{-1}{N}\right) \left(\frac{dQ}{l}\right) = \left(\frac{-1}{N}\right) \left(\frac{l}{dQ}\right) \\ &= \left(\frac{-1}{N}\right) \left(\frac{-1}{dQ}\right) \left(\frac{-l}{dQ}\right) \\ &= \left(\frac{-1}{s(d \mathrm{Nm}_{L/\mathbb{Q}} j_0)^{h'}}\right) \left(\frac{-1}{s' 3^{[L:\mathbb{Q}]} d \mathrm{Nm}_{L/\mathbb{Q}}(j_0)}\right) \left(\frac{-l}{3^{v_3(dQ)}}\right) \\ &= ss'(-1)^{[L:\mathbb{Q}] + v_3(dQ)}. \end{aligned}$$

If $[L : \mathbb{Q}] + v_3(dQ)$ is odd, let

$$j(a_1) \in \left(-\frac{16}{27}, -\frac{16}{27} + \min_{1 \leq i \leq r} \left|j_i + \frac{16}{27}\right|\right)$$

so that $(27j_i + 16)P_{-4l}(j_i) > 0$ for each $1 \leq i \leq r$. If $[L : \mathbb{Q}] + v_3(dQ)$ is even, by assumption let j_t be the minimal real conjugate of j_0 in $(-\frac{16}{27}, 0)$, and

$$j(a_1) \in (j_t, j_{t+1})$$

so that $(27j_i + 16)P_{-4l}(j_i) > 0$ for each $1 \leq i \leq r, i \neq t$ and $(27j_t + 16)P_{-4l}(j_t) < 0$. (Here we set $j_{r+1} = 0$.) In either case we have

$$\left(\frac{-4l}{N}\right) = -1$$

and there is a prime divisor p of N such that

$$\left(\frac{-4l}{p}\right) = -1.$$

The conditions on l and $(N, 6) = 1$ imply that $p \notin \text{Nm}_{L/\mathbb{Q}}(S)$ and $v_p(\text{Nm}_{L/\mathbb{Q}}(P_D(j_0))) = v_p(N) > 0$. If $l|N$, then since $l \notin \text{Nm}_{L/\mathbb{Q}}(S)$ by construction, we can choose $p = l$.

(2) Choose a prime l satisfying the conditions:

- $l \equiv 19 \pmod{24}$,
- $\left(\frac{q}{l}\right) = \left(\frac{-l}{q}\right) = 1$ for every prime $q \in \text{Nm}_{L/\mathbb{Q}}(S) \setminus \{2, 3\}$,
- $P_{-l}(x)$ has a real root $j(a_1)$ in a subinterval of $(0, \infty)$ to be specified later.

Let $D = -l$. By Lemma 3.6, Lemma 3.1, and [BG08b, Lemma 4],

$$P_{-l}(x) \equiv 1 \pmod{2},$$

$$P_{-l}(x) \equiv \pm x^{h'} \pmod{3},$$

$$P_{-l}(x) \equiv 3(27x + 16)S(x)^2 \pmod{l} \text{ for some } S(x) \in \mathbb{Z}[x].$$

Since $v_{\mathfrak{p}}(j_0) \geq 0$ for $\mathfrak{p}$ above 2 and $v_{\mathfrak{p}}(j_0) \leq 0$ for $\mathfrak{p}$ above 3, we can choose d such that $d \prod_{\sigma \in T} \sigma(j_0)$ is integral for any $T \subset \text{Hom}(L, \overline{\mathbb{Q}})$ and any prime dividing d lies in $\text{Nm}_{L/\mathbb{Q}}(S) \setminus \{2\}$, then $N, dQ \in \mathbb{N}$ and $(N, 6) = 1$. Suppose $l \nmid N$, then

$$\begin{aligned} \left(\frac{-l}{N}\right) &= \left(\frac{N}{l}\right) \\ &= \left(\frac{s(d \text{Nm}_{L/\mathbb{Q}}(3(27j_0 + 16)))}{l}\right) \\ &= \left(\frac{ss'3^{[L:\mathbb{Q}]}dQ}{l}\right) \\ &= ss'(-1)^{[L:\mathbb{Q}] + v_3(dQ) + v_2(Q)} \end{aligned}$$

Let j_t be the minimal real conjugate of j_0 in $(0, \infty)$ and n ($0 \leq n < r$) be the number of real conjugates of j_0 in $(-\frac{16}{27}, 0)$. If $[L : \mathbb{Q}] + v_3(dQ) + v_2(Q)$ is odd, let

$$j(a_1) \in \begin{cases} (j_t, j_{t+1}) & n \text{ is odd} \\ (0, j_t) & n \text{ is even} \end{cases}$$

so that $\text{Nm}_{L/\mathbb{Q}}(P_{-l}(j_0)) \text{Nm}_{L/\mathbb{Q}}(27j_0 + 16) > 0$. If $[L : \mathbb{Q}] + v_3(dQ) + v_2(Q)$ is even, let

$$j(a_1) \in \begin{cases} (j_t, j_{t+1}) & n \text{ is even} \\ (0, j_t) & n \text{ is odd} \end{cases}$$

so that $\text{Nm}_{L/\mathbb{Q}}(P_{-l}(j_0)) \text{Nm}_{L/\mathbb{Q}}(27j_0 + 16) < 0$. (Here we set $j_{r+1} = \infty$.) In either case we have

$$\left(\frac{-l}{N}\right) = -1$$

and there is a prime divisor p of N such that

$$\left(\frac{-l}{p}\right) = -1.$$

The conditions on l and $(N, 6) = 1$ imply that $p \notin \text{Nm}_{L/\mathbb{Q}}(S)$ and $v_p(\text{Nm}_{L/\mathbb{Q}}(P_D(j_0))) = v_p(N) > 0$. If $l|N$, then since $l \notin \text{Nm}_{L/\mathbb{Q}}(S)$ by construction, we can choose $p = l$.

(3) Choose a prime l satisfying the conditions:

- $l \equiv 1 \pmod{24}$,

- $\left(\frac{-3l}{q}\right) = 1$ for every prime $q \in \text{Nm}_{L/\mathbb{Q}}(S) \setminus \{2, 3\}$,
- the number of real conjugates of j_0 between the two real roots of $P_{-3l}(x)$ is odd, so that

$$\text{Nm}_{L/\mathbb{Q}}(P_{-3l}(j_0)) < 0.$$

Let $D = -3l$. By [BG08b, Lemma 4], the only possible odd prime power in $b_{h'}$ is 3 where 3 is a square modulo l . Then by Lemma 3.6 and Lemma 3.1,

$$\begin{aligned} P_{-3l}(x) &\equiv 1 \pmod{2}, \\ P_{-3l}(x) &\equiv \pm 1 \pmod{3}, \\ P_{-3l}(x) &\equiv S(x)^2 \pmod{l} \text{ for some } S(x) \in \mathbb{Z}[x]. \end{aligned}$$

Since $v_{\mathfrak{p}}(j_0) \geq 0$ for $\mathfrak{p}$ above 2, 3, we can choose d such that $d \prod_{\sigma \in T} \sigma(j_0)$ is integral for any $T \subset \text{Hom}(L, \overline{\mathbb{Q}})$ and any prime dividing d lies in $\text{Nm}_{L/\mathbb{Q}}(S) \setminus \{2, 3\}$, then $N \in \mathbb{N}$. Since $P_{-3l}(x)$ is constant modulo 6 and $[L : \mathbb{Q}]$ is even, we have

$$d^{h'} \text{Nm}_{L/\mathbb{Q}}(P_{-3l}(j_0)) \equiv 1 \pmod{6}.$$

Suppose $l \nmid P$, then

$$\left(\frac{-3l}{N}\right) = \left(\frac{N}{3l}\right) = \left(\frac{-1}{3l}\right) \left(\frac{d^{h'} \text{Nm}_{L/\mathbb{Q}}(P_{-3l}(j_0))}{3l}\right) = \left(\frac{-1}{3l}\right) = -1$$

since $P_{-3l}(x)$ is a square modulo l . There is a prime divisor p of N such that

$$\left(\frac{-3l}{p}\right) = -1.$$

The conditions on l and $(N, 6) = 1$ imply that $p \notin \text{Nm}_{L/\mathbb{Q}}(S)$ and $v_p(\text{Nm}_{L/\mathbb{Q}}(P_D(j_0))) = v_p(N) > 0$. If $l \mid N$, then since $l \notin \text{Nm}_{L/\mathbb{Q}}(S)$ by construction, we can choose $p = l$.

Remark 3.7. One can try to weaken the local conditions further by considering Heegner cycles with more general forms of discriminant D . For each prime p dividing D , the unpaired roots of $P_D(x)$ modulo p can be predicted by checking whether a maximal order in the definite quaternion algebra ramified at 2, 3, p contains two relevant CM orders whose trace-zero elements anticommute, as computed in [Jao03, Theorem 3.4.2]. One can then impose conditions on the number of prime divisors of D and the congruence class modulo 24 of the primes dividing D , so that $P_D(x)$ has no unpaired roots or $P_D(x)$ has a single unpaired root from the same elliptic point modulo each $p \mid D$. In addition, one imposes a congruence condition on D so that $\mathcal{P}_{D,\varepsilon}$ is disjoint from one of $\mathcal{P}_{-3,\varepsilon}$ and $\mathcal{P}_{-4,\varepsilon}$ by Lemma 3.5. The local conditions on j_0 are determined so that it avoids intersection with these $\mathcal{P}_D$ at $p = 2, 3$, and some real conjugate of j_0 and some real root of P_D lie in the subinterval corresponding to one geodesic. For D not of the form $-Np$ with $N = 1, 3, 4, 8, 12, 24$, it is possible that $P_D(x)$ has multiple real roots in each subinterval, but we still expect an equidistribution result.

4. PROOF OF THEOREM 1.3

In this section, we prove Theorem 1.3 by adapting the methods of [Che25] to the case in which the defining quaternion algebra is ramified at a finite place. In §4.1, we study the reduction of the CM cycles at good primes using special endomorphisms, paying particular attention to elliptic points with automorphisms of even order. In §4.2, we describe the real CM points on the relevant geodesics and apply Hecke equidistribution to control their positions at the two real places of F . In §4.3, we analyze the Čerednik–Drinfeld special fiber at $\mathfrak{p}_2$ and distinguish the reductions of CM points according to whether $\mathfrak{p}_2$ is ramified or inert in their CM fields. After recording the resulting local hypotheses in §4.4, we combine these ingredients in §4.5 to complete the proof.

Throughout this section, let $F = \mathbb{Q}(\sqrt{5})$ and B be the quaternion algebra over F ramified exactly at $\mathfrak{p}_2$ and at one real place of F . Recall that $\rho : F \hookrightarrow \mathbb{R}$ denotes the real place where B is split. Write $\mathcal{O}_F = \mathbb{Z}[\epsilon]$, where ϵ is the fundamental unit satisfying $\epsilon^2 - \epsilon - 1 = 0$ and $\rho(\epsilon) < 0$.

4.1. Integral model at good primes. Let p be an odd prime unramified in F and suppose that $B_{\mathfrak{p}}$ is split over $F_{\mathfrak{p}}$ for all $\mathfrak{p}|p$. Consider the integral model $\mathcal{S}_{K_p K^p}$ described in [Che25, §2]. Let $x \in \mathcal{S}_{K_p K^p}(k)$ be a closed point, $L(\mathcal{A}_x)$ be its space of special endomorphisms and $V(\mathcal{A}_x) = L(\mathcal{A}_x) \otimes_{\mathbb{Z}} \mathbb{Q}$. As a quadratic space over F , $V(\mathcal{A}_x)$ is positive definite at all real places and of dimension 3 with Hasse invariant $\epsilon(V(\mathcal{A}_x)_{F_v}) = 1$ for all finite primes $v \nmid 2p$, and $\epsilon(V(\mathcal{A}_x)_{F_v}) = -1$ for $v = \mathfrak{p}_2$. Moreover, if p is split in F , then $V(\mathcal{A}_x)$ has determinant 1 and the Hasse invariant of $V(\mathcal{A}_x)$ is -1 at $\mathfrak{p}$ over p corresponding to $\rho' : F \hookrightarrow W[p^{-1}]$, and 1 at the other $\mathfrak{p}$ over p , where $\rho' : F \hookrightarrow W[p^{-1}]$ satisfies $\rho = \iota \circ \rho'$ for the fixed embedding $\iota : \overline{W[p^{-1}]} \hookrightarrow \mathbb{C}$.

For an $\mathcal{S}_{K_p K^p}$ -scheme T , a point in $\mathcal{S}_{K_p K^p}(T)$ gives a triple $(\mathcal{A}, \lambda, \bar{\eta})$, where $(\mathcal{A}, \lambda : \mathcal{A} \rightarrow \mathcal{A}^{\vee})$ is a polarized abelian scheme over T up to isomorphism and $\bar{\eta} \in \Gamma(T, \underline{\text{Isom}}_G(H_{\mathbb{Z}^p}, \varprojlim_{p \nmid n} \mathcal{A}[n])/K^p)$ is a level structure. Two points $x_1, x_2 \in \mathcal{S}_{K_p K^p}(\mathbb{C})$, corresponding to $(\mathcal{A}_{x_1}, \lambda_1, \bar{\eta}_1), (\mathcal{A}_{x_2}, \lambda_2, \bar{\eta}_2)$, are identified in $\mathcal{S}_{K_0}$ if there is a prime-to- p quasi-isogeny $\mathcal{A}_{x_1} \rightarrow \mathcal{A}_{x_2}$ that preserves polarization up to $\mathbb{Z}_{(p)}^{\times}$, sends s_{α, B, x_1} to s_{α, B, x_2} and takes $\mathbf{L}_{\ell, x_1}$ to $\mathbf{L}_{\ell, x_2}$ for all $\ell \neq p$.

Proposition 4.6 of [Che25] remains valid with the same proof. When x has no automorphisms of even order, CM liftings of x to the cycle corresponding to $\mathcal{O}_{L_{\lambda}}$ occur in pairs.

We now turn to the points with even-order automorphisms. When $\mathfrak{p}_2$ is the only finite prime ramified in B , the proof of Lemma 4.5 in [Che25] can be modified to the present setting to obtain the following.

Lemma 4.1. *Let p be a rational prime that is totally split in F . Suppose $\mathcal{A}_x$ is supersingular and there is an order-2 automorphism $\alpha \in \text{SO}(V(\mathcal{A}_x), Q'_F)$ such that $\alpha(\mathbf{L}_{\text{cris}, x}) = \mathbf{L}_{\text{cris}, x}$ under the identification $L(\mathcal{A}_x) \otimes_{\mathbb{Z}} W \simeq \mathbf{L}_{\text{cris}, x}$ and $\alpha(\mathbf{L}_{\ell, x}) = \mathbf{L}_{\ell, x}$ under the identification $L(\mathcal{A}_x) \otimes_{\mathbb{Z}} \mathbb{Z}_{\ell} \simeq \mathbf{L}_{\ell, x}$ for all $\ell \neq p$, then there is $u \in L(\mathcal{A}_x)$ such that $Q'_F(u) = 1$ or 2 and $\alpha(u) = u$.*

Proof. The proof follows that of [Che25, Lemma 4.5], with the only modification being that there is no control on the parity of the valuation of $Q'_F(u') \in F_{\mathfrak{p}_2}$ for the constructed $u' \in V(\mathcal{A}_x)$, leading to the two possibilities for $Q'_F(u)$ after scaling. $\square$

Let p be a large enough rational prime that is totally split in F and λ be a totally positive prime of F above p corresponding to $\rho' : F \hookrightarrow W[p^{-1}]$. Let $x \in \mathcal{S}_{K_0}(k)$. Suppose $\mathcal{A}_x$ is supersingular, x has an even-order automorphism, and there is $e \in L(\mathcal{A}_x)$ with $Q'_F(e) = \lambda$. A slight modification of [Che25, Lemma 4.9(1)] shows that the special endomorphism $u \in L(\mathcal{A}_x)$ constructed in Lemma 4.1 is orthogonal to e . Since $V(\mathcal{A}_x)$ is a ternary orthogonal space of determinant 1 and Hasse invariant -1 at λ , if

$$\left(\frac{-1}{\lambda}\right) = 1,$$

then we cannot have $Q'_F(u) = 1$ in Lemma 4.1; if

$$\left(\frac{-2}{\lambda}\right) = 1,$$

then we cannot have $Q'_F(u) = 2$ in Lemma 4.1. In particular, when the totally negative prime $-\lambda$ is a square modulo 4 and a nonsquare modulo 8, then at λ the CM cycle intersects the order-2 elliptic point but not the order-4 elliptic point.

4.2. Real points. The unit ϵ satisfies $N_{F/\mathbb{Q}}(\epsilon) = \epsilon(1 - \epsilon) = -1$. The quaternion algebra

$$B \simeq \left(\frac{-\epsilon, -1}{F}\right),^4$$

and can be realized as

$$B \simeq \left\{ \begin{pmatrix} a & b \\ -b' & a' \end{pmatrix} : a, b \in F(\sqrt{-\epsilon}) \right\},$$

where $'$ denotes the nontrivial involution of $F(\sqrt{-\epsilon})/F$. Let

$$\mu = \begin{pmatrix} \sqrt{-\epsilon} & 0 \\ 0 & -\sqrt{-\epsilon} \end{pmatrix}, \nu = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \mu\nu = \begin{pmatrix} 0 & \sqrt{-\epsilon} \\ \sqrt{-\epsilon} & 0 \end{pmatrix}.$$

⁴This is the quaternion algebra split at all odd primes and at exactly one of the real places ρ_1 of F .

An explicit maximal order is

$$\mathcal{O} \simeq \mathcal{O}_F + \mathcal{O}_F \mu + \mathcal{O}_F \nu + \mathcal{O}_F \frac{1 + (2 - \epsilon)\mu + \nu + (2 - \epsilon)\mu\nu}{2}.$$

The element

$$w_2 = 1 + \nu$$

has reduced norm 2 and normalizes $\mathcal{O}$. It represents the $\mathfrak{p}_2$ -Atkin-Lehner element.

Let $\varphi : \mathcal{H} \rightarrow \Gamma_0 \backslash \mathcal{H}$ be the natural map. As in [Che25, 5.1], the action of complex conjugation on the Shimura curve $\Gamma_0 \backslash \mathcal{H}$ satisfies

$$\overline{\varphi(z)} = \varphi(\alpha(\bar{z})),$$

for any $\alpha \in N_{B^\times}(\mathcal{O})$ with $\rho(\text{nrd}(\alpha)) < 0$. For such α with $\text{trd}(\alpha) = 0$, if there is a CM point $z \in \mathcal{H}$, fixed by $v \in \mathcal{O}$ with $\text{trd}(v) = 0$ and $\text{nrd}(v) = \lambda$, lying on the geodesic Z_α , then since $\alpha v \alpha^{-1}(z) = \alpha v(\bar{z}) = \alpha(\bar{z}) = z$, we must have $\alpha v \alpha^{-1} = \bar{v} = -v$ ($\alpha v \alpha^{-1} \neq v$ as $v \notin F(\alpha)$), which implies $v \perp \alpha$. Consider $(B, N := \text{nrd})$ as a quadratic space over F , and let Λ be the orthogonal complement of $F(\alpha)$. The choice of α gives an optimal embedding of $R \hookrightarrow \mathcal{O}$, where R is an order of $F(\alpha)$ containing $\mathcal{O}_F[\alpha]$. The multiplication in the quaternion algebra gives $\Lambda \cap \mathcal{O}$ the structure of a module over R such that $N(aw) = \text{Nm}_{F(\alpha)/F}(a)N(w)$ for $a \in R, w \in \Lambda \cap \mathcal{O}$.

The field F has narrow class number 1. The prime $\mathfrak{p}_2 = 2\mathcal{O}_F$ ramifies in $F(\sqrt{-\epsilon})$ and $F(\sqrt{-2\epsilon})$, and both fields have class numbers 1. Consequently, as in [Che25, 5.1], we may assume the real points of $\tilde{\Gamma} \backslash \mathcal{H}$ are given by the image under $\varphi : \mathcal{H} \rightarrow \tilde{\Gamma} \backslash \mathcal{H}$ of the two geodesics

$$(4.2.1) \quad Z_\mu = \{iy : y > 0\},$$

$$(4.2.2) \quad Z_{\mu+\mu\nu} = \{x + iy : (x-1)^2 + y^2 = 2, y > 0\}.$$

Let $\lambda \in \mathcal{O}_F$ be totally positive and suppose there is a CM point fixed by $v \in \mathcal{O}$ with $\text{trd}(v) = 0$ and $\text{nrd}(v) = \lambda$. If the CM point lies on Z_μ , then

$$v = x\nu + y\mu\nu,$$

where $x, y \in \mathcal{O}_F$ satisfy the condition

$$\lambda = x^2 + \epsilon y^2 = \text{Nm}_{F(\sqrt{-\epsilon})/F}(x + y\sqrt{-\epsilon}).$$

If the CM point lies on $Z_{\mu+\mu\nu}$, then

$$v = x\nu + y(\mu - \mu\nu),$$

where $x, y \in \mathcal{O}_F$ satisfy the condition

$$\lambda = x^2 + 2\epsilon y^2 = \text{Nm}_{F(\sqrt{-2\epsilon})/F}(x + y\sqrt{-2\epsilon}).$$

Now assume λ is an odd prime satisfying (2.2.1), then λ is not represented by $x^2 + \epsilon y^2$, while λ is represented by $x^2 + 2\epsilon y^2$. Therefore there are no CM points with CM by $F(\sqrt{-\lambda})$ on the norm ϵ real geodesic. Put

$$S = \mathcal{O}_F[\mu + \mu\nu] \subset F(\sqrt{-2\epsilon}).$$

The relevant CM elements are exactly

$$v = a\nu, \quad a \in S, \quad N_{F(\sqrt{-2\epsilon})/F}(a) = \lambda.$$

If $a_1, a_2 \in S$ both have norm λ , then the ideals (a_1) and (a_2) are invertible prime ideals in S above (λ) . Hence

$$a_2 = ua_1 \quad \text{or} \quad a_2 = ua'_1 = u\nu a_1 \nu^{-1}$$

for some norm-one unit $u \in S^\times$, where $'$ denotes the nontrivial involution of $F(\sqrt{-2\epsilon})/F$. Note that $\eta := 2(1 - \epsilon) + (\epsilon - 2)(\mu + \mu\nu) \in S$ normalizes $\mathcal{O}$ and $\eta\nu\eta^{-1} = (5 + 2\sqrt{5} - (4 + 2\sqrt{5})(\mu + \mu\nu))\nu$, where $5 + 2\sqrt{5} - (4 + 2\sqrt{5})(\mu + \mu\nu)$ is a generator of the rank 1 group $S^{\times,1} := \{u \in S^\times : \text{Nm}_{F(\sqrt{-2\epsilon})/F}(u) = 1\} \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}$. Therefore, there is a unique real CM point

$$\varphi \left(-\frac{y\sqrt{-\rho(\epsilon)}}{x + y\sqrt{-\rho(\epsilon)}} + i \frac{\sqrt{\lambda}}{|x + y\sqrt{-\rho(\epsilon)}|} \right)$$

corresponding to an optimal embedding $\mathcal{O}_{L_\lambda} \hookrightarrow \mathcal{O}$, where $x, y \in \mathcal{O}_F$ satisfy $\lambda = x^2 + 2\epsilon y^2$. This real CM point lies on the image of the geodesic $Z_{\mu+\mu\nu} = \{x + iy : (x-1)^2 + y^2 = 2, y > 0\}$. Moreover, by Hecke's equidistribution, for any open segment of the geodesic and any subclass of the congruence class specified in (2.2.1), there exist infinitely many primes λ in the sub-congruence class such that the unique real CM point corresponding to $\mathcal{O}_{L_\lambda} \hookrightarrow \mathcal{O}$ lies in the image of the open segment.

For the application we should apply the preceding construction simultaneously to the two conjugate Shimura curves. The field $F(\sqrt{-2\epsilon}, \sqrt{-2(1-\epsilon)})$ has class number 1. Applying the construction of [Che25, Lemma 5.4] gives a Hecke character from $\mathbb{A}_F^\times / F(\sqrt{-2\epsilon}, \sqrt{-2(1-\epsilon)})^\times$ onto the product of the two real geodesics. Hecke equidistribution gives infinitely many totally positive primes satisfying the prescribed finite congruences and the two independent conditions at the two real places.

4.3. Integral model and reduction of CM points at the bad prime. Consider the Čerednik–Drinfeld integral model at the ramified prime $\mathfrak{p}_2$. Recall that B is ramified at $\mathfrak{p}_2$ and at the real place ρ_2 . Let B' be the definite quaternion algebra obtained from B by interchanging local invariants at $\mathfrak{p}_2$ and at the split real place ρ_1 . Thus B' is ramified exactly at the two real places and is unramified at every finite place. Let $\mathcal{O}' \subset B'$ be a maximal order.

We follow Kurihara [Kur79] to compute the quotient of the Bruhat–Tits tree, which is dual to the graph of the geometric special fiber. Let $\mathcal{T}$ denote the Bruhat–Tits tree of $\mathrm{PGL}_2(F_{\mathfrak{p}_2})$, and put

$$\Gamma'_0 = \mathcal{O}'[2^{-1}]^\times / \mathcal{O}_F[2^{-1}]^\times \subset \mathrm{PGL}_2(F_{\mathfrak{p}_2}).$$

The vertex set of $\Gamma'_0 \backslash \mathcal{T}$ has cardinality equal to the class number of this definite quaternion algebra B' . In the present case this class number is 1. Since the Shimura curve $\mathrm{Sh}_{K_0}(G_0, \mathcal{H}^\pm)$ has genus 0, by [Kur79, Proposition 3-2], the graph $\Gamma'_0 \backslash \mathcal{T}$ only has self-inverse edges. The vertex v in $\mathcal{T}$ corresponding to $\mathcal{O}'_{\mathfrak{p}_2}$ has stabilizer $\Gamma'_{0,v} \simeq \mathcal{O}'^\times / \mathcal{O}_F^\times$ in Γ'_0 . Note that the reduction map $\mathcal{O}'^\times / \mathcal{O}_F^\times \rightarrow \mathrm{PGL}_2(\mathbb{F}_4)$ is an isomorphism, so $\Gamma'_{0,v}$ acts transitively on the $4 + 1 = 5$ edges around v .

Consequently, the geometric special fiber of the Čerednik–Drinfeld model at $\mathfrak{p}_2$ is $\mathbb{P}^1$ and there is a distinguished smooth point corresponding to the image of edges in $\Gamma'_0 \backslash \mathcal{T}$.

In the setting of a Shimura curve with Čerednik–Drinfeld reduction at a finite ramified place v , the reduction of a CM point corresponds to the image of a singular point in the special fiber (edge in the Bruhat–Tits tree) precisely in the ramified case (see [Sae24, Proposition 2.17] for a proof). Applying this at $v = \mathfrak{p}_2$, the CM points for which $\mathfrak{p}_2$ ramifies in L_λ and the CM points for which $\mathfrak{p}_2$ is inert in L_λ have disjoint reductions on the Čerednik–Drinfeld special fiber.

4.4. Local conditions.

4.4.1. Coordinate function. The order-4 elliptic point is F -rational and is CM by $F(\sqrt{-1})$. Choose an integral coordinate on the coarse $\mathbb{P}^1$ -model by applying an element of $\mathrm{PGL}_2(\mathcal{O}_F)$ that sends this section to ∞ . Denote the resulting coordinate by j .

4.4.2. Local condition at the real places. At the real places ρ_1, ρ_2 , we impose at least one of the following conditions on the coordinate j_0 of the fixed abelian variety A under consideration:

- (1) $\rho_1(j_0)$ lies on the geodesic between the order-2 elliptic point and order-4 elliptic point defined by $\alpha \in N_{B^\times}(\mathcal{O})$ with $\mathrm{trd}(\alpha) = 0$ and $\mathrm{nrd}(\alpha) = 2\epsilon$;
- (2) $\rho_2(j_0)$ lies on the geodesic between the order-2 elliptic point and order-4 elliptic point defined by $\alpha \in N_{B^\times}(\mathcal{O})$ with $\mathrm{trd}(\alpha) = 0$ and $\mathrm{nrd}(\alpha) = 2(1 - \epsilon)$.

4.4.3. Local condition at the bad prime. With the choice of coordinate j , the condition $v_{\mathfrak{p}_2}(j_0) < 0$ is equivalent to j_0 specializing to the distinguished point corresponding to the image of edges in the Bruhat–Tits tree. We impose this local condition on the coordinate of the fixed abelian variety A under consideration.

4.5. Proof of main theorem. Let j_0 be the coordinate of the point on the Shimura curve $\text{Sh}_{K_0}(G_0, \mathcal{H}^\pm)$ given by the fixed abelian variety A . The result is known when A has complex multiplication (see [Che25, 3.10]), so we may assume that j_0 is a non-CM point on the Shimura curve. Assume $j_0 \in F$ satisfies the local conditions 4.4.3 and 4.4.2.

Let j_2 be the coordinate of the order-2 elliptic point. Recall that the coordinate function is chosen such that ∞ is the order-4 elliptic point. Let S' be a finite set of rational primes such that $S' \supset \{2, 5\} \cup \{p : v_{\mathfrak{p}}(j_0) \neq 0 \text{ or } v_{\mathfrak{p}}(j_2) \neq 0 \text{ or } v_{\mathfrak{p}}(j_2 - j_0) \neq 0 \text{ for some prime } \mathfrak{p} \text{ of } F \text{ above } p\}$. Let S be the set of prime ideals of F above the primes in S' . Let $d_0 \in \mathcal{O}_F$ be a totally positive denominator of j_0 such that $d_0 j_0 \in \mathcal{O}_F$ and $(d_0, d_0 j_0) = 1$. In particular, all primes dividing d_0 lie in S .

For a large enough totally positive prime $\lambda \notin S$ of F satisfying (2.2.1), let $\alpha_1, \dots, \alpha_h$ be the j -coordinates of the CM points in the CM cycle corresponding to optimal embeddings $\mathcal{O}_{F(\sqrt{-\lambda})} \hookrightarrow \mathcal{O}$, where $h = h(F(\sqrt{-\lambda}))$ is odd, and define the polynomial

$$P_\lambda(x) := \prod_{l=1}^h (x - \alpha_l).$$

By construction, the CM cycle has reflex field L_λ and it is stable under complex conjugation. Thus $\text{Gal}(\overline{F}/F)$ permutes $\alpha_1, \dots, \alpha_h$, and $P_\lambda(x) \in F[x]$. Let $b_\lambda \in \mathcal{O}_F$ be the totally positive denominator of $P_\lambda(x)$ such that $b_\lambda P_\lambda(x)$ is primitive. Note that $\alpha_1, \dots, \alpha_h \in H_{F(\sqrt{-\lambda})}$, where $H_{F(\sqrt{-\lambda})}$ is the Hilbert class field of $F(\sqrt{-\lambda})$. The principal prime ideal $(\sqrt{-\lambda})$ of $\mathcal{O}_{F(\sqrt{-\lambda})}$ splits completely in $H := H_{\mathcal{O}_{F(\sqrt{-\lambda})}}$. Let $\mathfrak{l}$ be a prime of H above λ . By §4.1, the choice (2.2.1) implies that $v_{\mathfrak{l}}(\alpha_k) \geq 0$ for all $k = 1, \dots, h$, then $\lambda \nmid b_\lambda$. By §4.3, since $\mathfrak{p}_2$ is ramified in $F(\sqrt{-1})$ and inert in $F(\sqrt{-\lambda})$, we have $v_{\mathfrak{p}_2}(\alpha_k) \geq 0$ for all $k = 1, \dots, h$, then $2 \nmid b_\lambda$.

As explained in §4.1, the roots of $P_\lambda(x)$ modulo $\mathfrak{l}$ occur in pairs, except for those congruent to j_2 . From the assumption that $\lambda \notin S$, we have $v_{\mathfrak{l}}(d_0) = 0$, then modulo $\mathfrak{l}$, we have

$$(4.5.1) \quad d_0^h b_\lambda P_\lambda(j_0) \equiv d_0 b_\lambda (j_0 - j_2) \cdot \text{square} \pmod{\mathfrak{l}}.$$

Write $d_0(j_0 - j_2) = (-1)^s \epsilon^t 2^m n$ where $s, t \in \{0, 1\}$, $m \in \mathbb{Z}$, and $n \in F$ is totally positive with $v_{\mathfrak{p}_2}(n) = 0$. The assumption (2.2.1) implies

$$\left(\frac{(-1)^s \epsilon^t 2^m}{\lambda} \right) = (-1)^{t+m}$$

By the equidistribution result discussed in §4.2, there exists a totally positive prime $\lambda \notin S$ of F such that

- (1) $-\lambda$ is a square modulo 4 and not a square modulo 8;
- (2) $-\lambda$ is a square modulo all odd primes in S ;
- (3) If $t + m$ is even, then $P_\lambda(j_0)$ is positive at one real place and negative at the other; if $t + m$ is odd, then $P_\lambda(j_0)$ is totally positive or totally negative.

If λ divides $d_0^h b_\lambda P_\lambda(j_0)$, then j_0 is a root of $b_\lambda P_\lambda(x)$ modulo some prime above λ , where the prime (λ) is ramified in $F(\sqrt{-\lambda})$. Thus, the section defined by j_0 meets the closure of the CM cycle defined by L_λ at a prime above $(\lambda) \notin S$, where the corresponding CM Kuga-Satake abelian variety has supersingular reduction. Assume λ does not divide $d_0^h b_\lambda P_\lambda(j_0)$, then

$$\begin{aligned} \left(\frac{-\lambda}{d_0^h b_\lambda P_\lambda(j_0)} \right) &= (-1)^{\frac{\text{sgn}(\rho_1(P_\lambda(j_0))) + \text{sgn}(\rho_2(P_\lambda(j_0)))}{2} - 1} \left(\frac{d_0^h b_\lambda P_\lambda(j_0)}{\lambda} \right) \\ &= (-1)^{\frac{\text{sgn}(\rho_1(P_\lambda(j_0))) + \text{sgn}(\rho_2(P_\lambda(j_0)))}{2} - 1} \left(\frac{d_0 b_\lambda (j_0 - j_2)}{\lambda} \right) \\ &= - \left(\frac{b_\lambda n}{\lambda} \right) \\ &= - \left(\frac{-\lambda}{b_\lambda n} \right) \\ &= -1. \end{aligned}$$

For the last equality, note that for $\mathfrak{q} \notin \{(2), (\sqrt{5}), (\lambda)\}$, the order-4 stabilizer of ∞ implies that $v_{\mathfrak{q}}(b_\lambda)$ is divisible by 4 (see for instance [Che25, Lemma 2.5(2)]; only its evenness is used). Thus, there is a prime $\mathfrak{p}$

with $v_{\mathfrak{p}}(d_0^h b_\lambda P_\lambda(j_0))$ odd such that $\left(\frac{-\lambda}{\mathfrak{p}}\right) = -1$. Note that $\mathfrak{p} \neq \mathfrak{p}_2$ since $v_{\mathfrak{p}_2}(j_0) < 0$ and $v_{\mathfrak{p}_2}(\alpha_k) \geq 0$ for all $k = 1, \dots, h$. Moreover, $\mathfrak{p} \notin S$ since $\left(\frac{-\lambda}{\mathfrak{q}}\right) = 1$ for every odd prime $\mathfrak{q}$ in S . Then $v_{\mathfrak{p}}(b_\lambda)$ is even and j_0 is a root of $b_\lambda P_\lambda(x)$ modulo some prime above $\mathfrak{p}$. Therefore, the section defined by j_0 meets the closure of the CM cycle defined by L_λ at a prime above $\mathfrak{p} \notin S$, where the corresponding CM Kuga–Satake abelian variety has supersingular reduction.

REFERENCES

- [BC91] J.-F. Boutot and H. Carayol, *Uniformisation p -adique des courbes de Shimura: les théorèmes de čerednik et de Drinfeld*, 1991, pp. 7, 45–158. Courbes modulaires et courbes de Shimura (Orsay, 1987/1988). MR1141456
- [BG08a] Srinath Baba and Håkan Granath, *Genus 2 curves with quaternionic multiplication*, Canad. J. Math. **60** (2008), no. 4, 734–757. MR2423455
- [BG08b] ———, *Primes of superspecial reduction for QM abelian surfaces*, Bull. Lond. Math. Soc. **40** (2008), no. 2, 311–318. MR2414789
- [CH88] P. E. Conner and J. Hurrelbrink, *Class number parity*, Series in Pure Mathematics, vol. 8, World Scientific Publishing Co., Singapore, 1988. MR963648
- [Che25] Fangu Chen, *Infinitely many supersingular primes for some Mumford’s abelian fourfolds*, 2025.
- [Cla03] Pete L. Clark, *Rational points on Atkin–Lehner quotients of Shimura curves*, Ph.D. Thesis, 2003 (English). Copyright - Database copyright ProQuest LLC; ProQuest does not claim copyright in the individual underlying works; Last updated - 2023-03-04.
- [Elk87] Noam D. Elkies, *The existence of infinitely many supersingular primes for every elliptic curve over $\mathbf{Q}$* , Invent. Math. **89** (1987), no. 3, 561–567. MR903384
- [Elk89] ———, *Supersingular primes for elliptic curves over real number fields*, Compositio Math. **72** (1989), no. 2, 165–172. MR1030140
- [Elk98] ———, *Shimura curve computations*, Algorithmic number theory (Portland, OR, 1998), 1998, pp. 1–47. MR1726059
- [Jao03] David Yen Jao, *Supersingular primes for rational points on modular curves*, ProQuest LLC, Ann Arbor, MI, 2003. Thesis (Ph.D.)—Harvard University. MR2704678
- [KR08] Kevin Keating and David P. Roberts, *Intersection numbers of Heegner divisors on Shimura curves*, Pure Appl. Math. Q. **4** (2008), no. 4, 1165–1204. MR2441697
- [Kur79] Akira Kurihara, *On some examples of equations defining Shimura curves and the Mumford uniformization*, J. Fac. Sci. Univ. Tokyo Sect. IA Math. **25** (1979), no. 3, 277–300. MR523989
- [Lan94] Serge Lang, *Algebraic number theory*, Second, Graduate Texts in Mathematics, vol. 110, Springer-Verlag, New York, 1994. MR1282723
- [LMPT25] Wanlin Li, Elena Mantovan, Rachel Pries, and Yunqing Tang, *Infinitely many primes of basic reduction for some abelian fourfolds*, 2025.
- [Mum72] David Mumford, *An analytic construction of degenerating curves over complete local rings*, Compositio Math. **24** (1972), 129–174. MR352105
- [Rib89] Kenneth A. Ribet, *Bimodules and abelian surfaces*, Algebraic number theory, 1989, pp. 359–407. MR1097624
- [Rot04] Victor Rotger, *Modular Shimura varieties and forgetful maps*, Trans. Amer. Math. Soc. **356** (2004), no. 4, 1535–1550. MR2034317
- [Sad04] Marat Sadykov, *Two results in the arithmetic of Shimura curves*, ProQuest LLC, Ann Arbor, MI, 2004. Thesis (Ph.D.)—Columbia University. MR2705896
- [Sae24] Francesco Maria Saettone, *Equidistribution of CM points on a Shimura curve modulo a ramified prime*, 2024.
- [Tak77] Kisao Takeuchi, *Commensurability classes of arithmetic triangle groups*, J. Fac. Sci. Univ. Tokyo Sect. IA Math. **24** (1977), no. 1, 201–212. MR463116
- [vG08] Bert van Geemen, *Real multiplication on K3 surfaces and Kuga–Satake varieties*, Michigan Math. J. **56** (2008), no. 2, 375–399. MR2492400
- [Voi09] John Voight, *Shimura curves of genus at most two*, Math. Comp. **78** (2009), no. 266, 1155–1172. MR2476577